

Reproduction of EpiModX, Comparison to CNN+MoE, and CNN+BLSTM Architectures for Histone Modifications Prediction in AD and AML Datasets

Introduction

Epigenetic modifications, such as histone modifications and DNA methylation, have been shown to contribute significantly to the development of various diseases such as Alzheimer's Disease (AD) and Acute Myeloid Leukemia (AML). In approximately 45% of AML cases, for example, patients lack visible chromosomal abnormalities, but there are still defects detected at the molecular level, meaning that a significant portion of AML cases are largely the result of epigenetic factors. Much of the variations causing these changes also occur in the non-coding region of the genome, making it hard to determine how specifically they contribute to disease states, especially with models previously created to solve this problem. Therefore, this research dives more deeply into the question of how to model the effects of disease-associated histone modifications on disease progression.

Previous research on the topic of epigenetic prediction largely fits into two categories: variant effect prediction models and histone-specific predictions. Variant effect prediction models, such as DeepSEA and ExPecto, predict the effects of sequence variants on a variety of chromatin features. These models are able to predict DNA patterns from large-scale chromatin data, providing the understanding of epigenetic effects on the regulatory sequence, but they are not designed to be disease-conditioned. Histone-specific predictors, on the other hand, are designed to identify post-translational modification sites in histone tails. These models, including DeepHistone and Histone-Net, are able to predict the presence of histone modification marks, which are generally associated with gene regulatory activity, but they are not predictive across disease contexts.

Therefore, this previous research lacks the ability to predict histone modification patterns related to a specific disease, an important part of understanding pathogenic mechanisms that EpiModX attempts to learn. In learning these patterns, however, there are the challenges of sample size constraints and modeling the disease-contextual regulatory grammar.

EpiModX overcomes these limitations through the use of a large language model (LLM)-based framework that is designed for specific disease-contextual predictions. With the rise in development of large-scale DNA language models, genomic grammatical structures, syntactic patterns, and semantic relationships can be learned from sequence data. In the context of EpiModX, these LLMs allow the model to understand meaningful patterns, even when working with limited epigenetic data. This LLM is then combined with a Mixture of Experts (MoE) module that is able to capture disease-specific epigenetic variations in addition to the general rules of gene regulation.

While EpiModX shows promising results, we hope to address the potential shortcomings of generalizability and relative contribution of the LLM and MoE architecture in this study. The

original study tested this model on AD, but it is not addressed if this model can be generalized to strongly predict disease-specific histone modifications for other diseases. Thus, our extension of training and testing EpiModX to acute myeloid leukemia data aims to highlight the performance of EpiModX in a different disease context.

Moreover, the study emphasizes the value of the DNA language model as well as the Mixture of Experts in the model architecture. However, these are both very computationally expensive, which is a potential flaw in the model. Our partial ablation studies with reduced models (CNN + BLSTM, CNN + MoE) aims to investigate the necessity and effects of each model in producing robust predictions. Through these extensions, we will be able to understand the true performance of EpiModX and its feasibility for further development.

Methods

EpiModX is a tri-component architecture consisting of: 1) an DNA Large Language Model which extracts learned features from raw DNA sequence, 2) Convolutional Neural Network blocks which extracts genomic features, and 3) a Mixture of Experts (MoE) model which is designed to predict disease-specific histone modifications. We compare results of their original model with our replication, as well as a partial ablation study with smaller models: CNN+MoE and CNN+BLSTM to hone in on the effects of incorporating the biologically informed LLM as a pre-processing step for raw DNA sequence. We accomplish this through direct comparison of model output metrics such as Accuracy (ACC), F1-score, AUROC, and AUPRC.

1. Full Model Reproduction

This model was trained on ChIP-seq data for histones (H3K27ac, H3K4me3, and H3K27me3) from the ENCODE database, excluding regions from blacklists and sequences with unresolved bases. To integrate datasets, overlapping sequences were combined, centering on the peak and extending to 2048bp (decreased from 4096bp due to GPU memory limitations) to capture regulatory contexts.

Large-Language Model:

Recent advances in DNA language models have made it possible to consider disease-specific prediction of histone modification by learning the “grammatical structure” of sequence data. The DNA language model employed in the EpiModX framework is Caduceus, built on the long-range Mamba architecture and trained on the human genome, and processes raw DNA sequence to create embeddings that capture the aforementioned complex genetic information. The original study initializes Caduceus using pretrained weights from Hugging Face. This LLM encodes each DNA sequence into a 16-dimensional numerical feature vector that is later passed to the Convolutional Neural Network.

Convolutional Blocks:

A deep neural network module with four convolutional neural network layers followed by max pooling layers was used to reduce spatial dimensions and aggregate information.

Mixture of Experts:

To identify the difference between disease progression groups, the model uses a mixture of experts (MoE). This MoE module contains two MoE attention blocks, customised to capture disease-specific information, and two MoE MLP blocks. Each MoE block contains 16 experts along with five disease-specific routing networks, for each of the five disease groups. The output of the MoE attention layer is made up of the Noisy Routing network and the expert attention head's disease-specific embedding:

$$y = \sum_{k=1}^N G_i^k(x) \cdot E_k(x + e_i)$$

Where i is the disease group, G_i^k is the Noisy Routing network, e_i represents the disease-specific embedding, and E_k denotes the expert (the attention head). Finally, the model had five specialized output heads for each disease group.

Model training:

This model was implemented in Pytorch and employs early stopping, with a stopping criterion of 5 epochs. The loss function is the sum of binary cross-entropy loss and the weighted loss of the MoE module (0.5 for this study). The dataset was split into training, validation, and testing sets based on the chromosome.

In the original paper, the model was trained on a NVIDIA A100 GPU with a batch size of 8. However, due to our GPU limitations with the SCC and time constraints on the project, we sped up the training process through data parallelism across two GPUs with a compute capability of at least 8.0. Using PyTorch's Distributed Data Parallelism (DDP), we are able to process more data per step compared to the original single-GPU training. We used DistributedSampler to ensure each GPU gets a disjoint subset of the training data, essentially doubling the batch size. We wrap the model in DDP so that each GPU keeps track of all of the model's weights and after each backwards pass, gradients are averaged across all GPUs so that all models are able to stay in sync. Finally, when evaluating the model, the GPUs contribute their data to a single source, computing metrics across the entire model. Through this data parallelism, we were able to speed up the training of these models slightly, but were still limited by the number of GPUs we were allowed to request on the SCC.

In order to run this model and all subsequent extensions, detailed instructions are found in our GitHub code repository: <https://github.com/alicehan8/DS596FinalProject>.

Model testing:

For the most part, we were able to make use of the testing implementation of the original study. However, since we implemented parallelization in our model using DDP, this required us

to alter the testing of the model to account for this. A key challenge was resolving checkpoint keys saved during multi-GPU training with single-GPU inference, which required stripping DataParallel prefixes and remapping renamed layer attributes before loading model weights. For example, in the state dictionary of the model, DDP appends “module” before each key, whereas the original testing code provided by the study does not include that. To handle this, we first processed all the keys of the model’s state dictionary before testing. Moreover, some layers were renamed during development that require renaming for consistency. We adapted the testing file to support both AD and AML data. For AD testing, the model outputs predictions across 22 tasks categorized into 6 classes for three different disease states. For AML testing however, this was collapsed into just one binary task, so changes in the testing code were made for this too.

As for metrics computed AUC, F1, accuracy, and AUPRC were calculated then averaged for per task metrics for AD, and for AML, these were computed as global metrics since there is only one binary task. These metrics were calculated in a separate file we created as the study’s test only produced predictions and labels.

2. Ablation Study

We include two model architectures here for ablation study.

CNN+MoE:

To isolate the effects of the LLM in the pipeline, we conducted a partial ablation study with just the CNN+MoE components to compare differences in prediction when incorporating the grammatical structure from the DNA language model. In other words, the CNN+MoE ablation tests whether disease-specific performance comes from the MoE mechanism or the pretrained language model.

The CNN blocks are identical to those used in the original paper. In fact, the original codebank gives the option to withhold the LLM backbone in the pipeline. Specifically, the CNN portion maintains the 4-block convolutional encoder, kernel size = 5, max-pooling operation, etc.; however, the CNN here acts as a traditional feature extractor from raw DNA sequence alone — learning motif-esque regulatory patterns

The MoE component also stays the exact same as the full model — preserving the ability of the model to leverage disease-specific experts and retain epigenetic differences across various contexts.

CNN+BLSTM:

We chose CNN+BLSTM as our alternative model because its performance ranked 3rd in the paper, slightly lower than LLM+MoE and LLM+BLSTM. By contrast, its training should be way cheaper than the EpiModX model for not using LLM blocks. CNN+BLSTM also acts as another layer of comparison to CNN+MoE model, to justify whether the MoE is essential for learning.

The CNN block is identical to the one used in the EpiModX model. The input is first processed by a 4-block convolutional encoder that progressively extracts hierarchical sequence motifs. Each block consists of a 1D convolution with kernel size 5 and same-padding, followed by a ReLU activation and a max-pooling operation with stride 2 that halves the sequence length. The channel dimensions follow the progression 4-> 64 -> 128 -> 256 -> 128. After 4 pooling operations, the 4096bp input is reduced to a 256-position sequence of 128D feature vectors.

The input is from the output of CNN block. The BLSTM network has a hidden size of 64 per direction, yielding a 128-dimensional hidden state at each position. Dropout rate is set as 0.3 between 2 LSTM layers. The recurrent encoder produces a context-aware sequence of 256*128 hidden states.

A global mean-pooling operation collapses the 256-position hidden state sequence into a single 126-dimensional vector summarizing the entire window. This vector is passed through a layerNorm followed by a single linear projection to produce 22 output logits, one per patient. All 22 patients are predicted simultaneously from the shared representation. During training, each logit is independently supervised with binary cross-entropy loss against that patient's binary histone modification label for the corresponding genomic bin.

$$L = -\frac{1}{NP} \sum_{n=1}^N \sum_{p=1}^P [y_{n,p} \log \hat{y}_{n,p} + (1 - y_{n,p}) \log(1 - \hat{y}_{n,p})]$$

The AML generalization dataset is identical to the AD dataset. The only structural difference lies in the output head: the AML model changes the final linear layer to fit 57 outputs for H3K27me3 and H3K4me3 (38 AML and 19 Healthy), and 50 outputs for H3K27me3 (32 AML and 18 Healthy). The patient grouping structure is also different. AML cohort uses a binary grouping (AML vs. Healthy).

3. AML Database Extension

To evaluate the cross-disease generalization of EpiModX as well as the reduced models, we edited and trained the models on Acute myeloid leukemia (AML), one of the most epigenetically altered cancers. We constructed an independent epigenomic dataset pairing AML with healthy myeloid controls, both drawn from the Blueprint Epigenome project accessed via IHEC data portal to minimize batch effect.

AML samples were identified from the IHEC Blueprint metadata (32 samples) containing all histone modifications of interest (H3K4me3, H3K27me3, H3K27ac). Healthy myeloid controls are selected along the myeloid differentiation axis relevant to AML pathology with filter “disease = Healthy”. 53 unique donors out of 608 samples with 18-19 donors per histone modification are selected (H3K4me3 19; H3K27me3 18; H3K27ac 19).

In order to run these models on the AML data, they needed to first be modified to be a single binary prediction task (AML or not) rather than predicting across five tasks. Additionally, the loss function is calculated as direct binary cross entropy rather than being balanced across all tasks. Through these changes, we were able to train the EpiModX model on AML data to test how the model generalizes across different datasets.

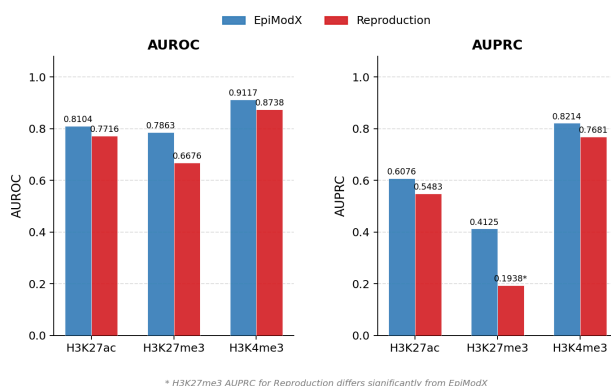
Results

In the original study, EpiModX demonstrated robust performance in prediction of disease-specific histone modifications, also demonstrating stronger performance than the existing state of the art methods. While these state of the art methods lack specific disease context, EpiModX was able to use disease context in its predictions. Aside from strong prediction performance, EpiModX also is able to prioritize disease-related genetic variants, specifically Alzheimer’s disease, which has a meaningful impact for biological discovery.

More specifically, EpiModX achieved AUROC scores of 0.8104 for H3K27ac, 0.9117 for H3K4me3, and 0.7863 for H3K27me3 on average, which was a significant improvement from the average AUROCs of the existing Expecto model. We can then compare these metrics to the metrics obtained in our team’s reproduction below.

1. AD Dataset Reproduction and Comparison:

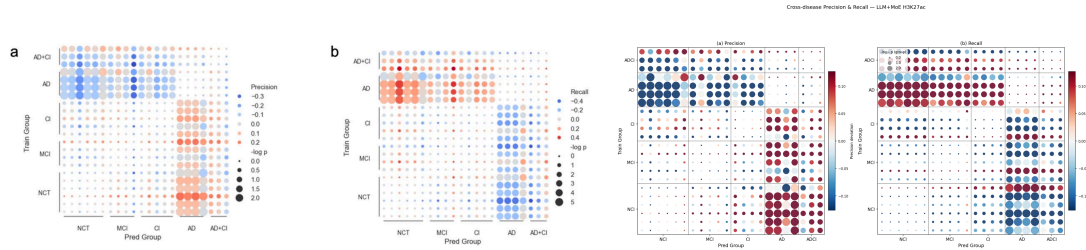
Full-model Reproduction:



Our full model reproductions achieve AUROC and AUPRC values that closely match those reported in the original study, which consistently preserve the same relative performance trends across the three histone modifications. This alignment suggests that the reproduced models successfully conduct the class prediction.

In addition to evaluating the AUROC and AUPRC of our reproduction, we also compared our accuracy and precision against the study. For each individual histone modification, we independently trained our parallelized model using the same datasets as the study. In all figures, the dot size represents $-\log_{10} P$ value and the color represents performance deviation in cross-disease analysis. Across all three modifications, we found that our reproduction results are closely aligned with the original study, as detailed below.

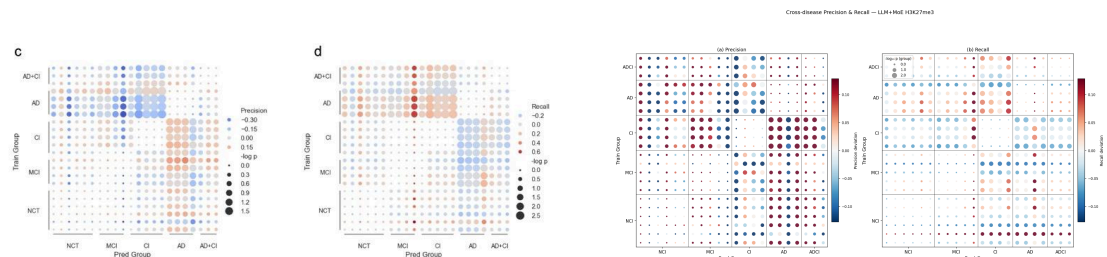
H3K27ac



Precision and recall metrics of H3K27ac across disease groups

The original figure (left) exhibits lower precision but higher recall for acetylation site predictions in the NCI (no cognitive impairment) group. This demonstrates potential biological bias in which AD-derived sequences are more frequently classified as acetylation sites than NCI-derived sequences. Comparing the original figure with our reproduced figure (right), the overall structure and distribution of precision and recall values are consistent, indicating that our reproduction makes similar predictions and even exhibits the same biological biases as the original model. However, there are small differences between the original and reproduced figures, where the precision is higher, recall lower, or certain regions have lower intensity, which may be due to differences in model training or randomness.

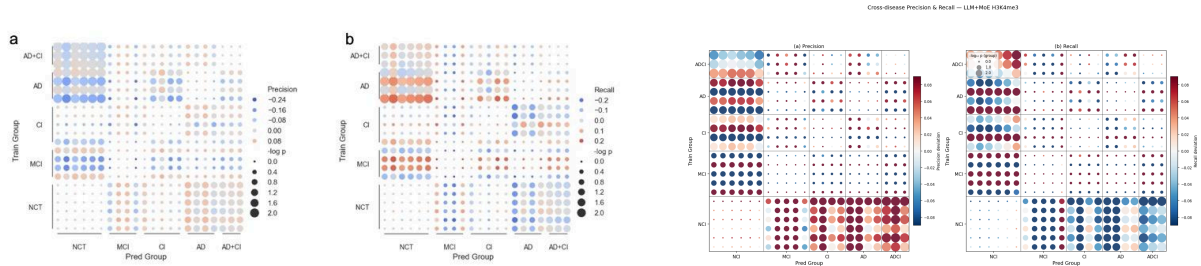
H3K27me3



Precision and recall metrics of H3K27me3 across disease groups

For H3K27me3, there is noticeable asymmetry between precision and recall. In the original figure (left) this is exhibited by the AD and AD+CI groups, which exhibit low precision but high recall when predicting NCI, MCI, and CI groups. Between the original figure and the reproduced figure (right), there is consistent block structure, indicating that our reproduction captures similar biological signals as the original study. H3K27me3 histone modification may be transferable across disease stages, however the low precision indicates a potential tendency to over-predict.

H3K4me3



Precision and recall metrics of H3K4me3 across disease groups

For H3K4me3, the NCI group shows the greatest asymmetry relative to the other groups, with high recall but low precision. For disease groups, they are able to predict other disease stages with high precision. Between the original (left) and reproduced (right) figures, the overall knowledge transfer structure is produced, however the reproduced figure exhibits stronger precision for disease group predictions.

2. Models Comparison in AD Dataset

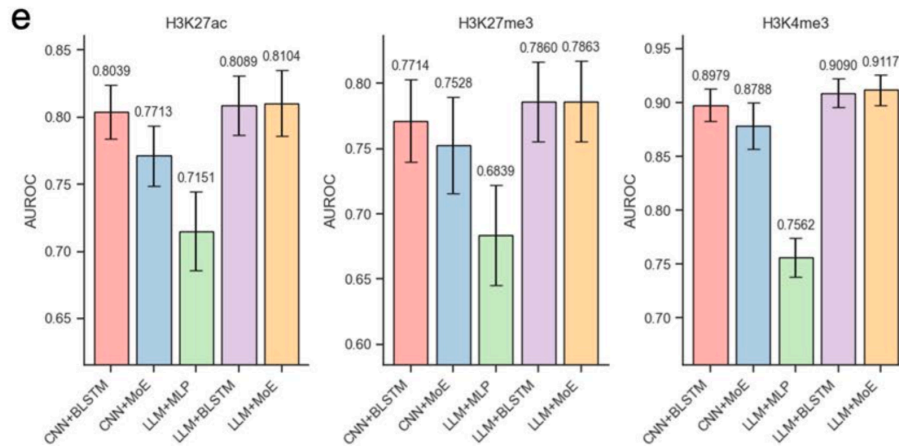
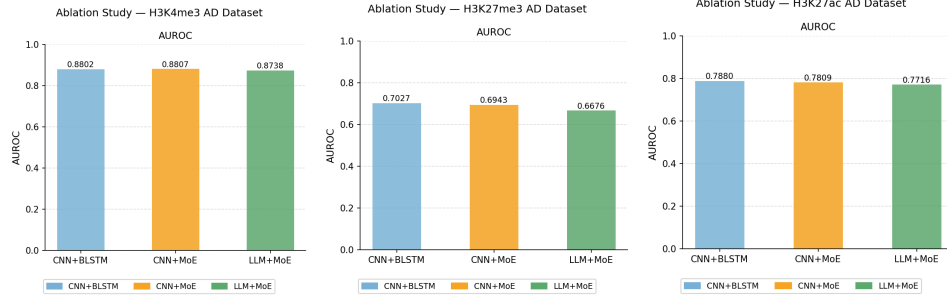


Fig.2e in the EpiModX original study

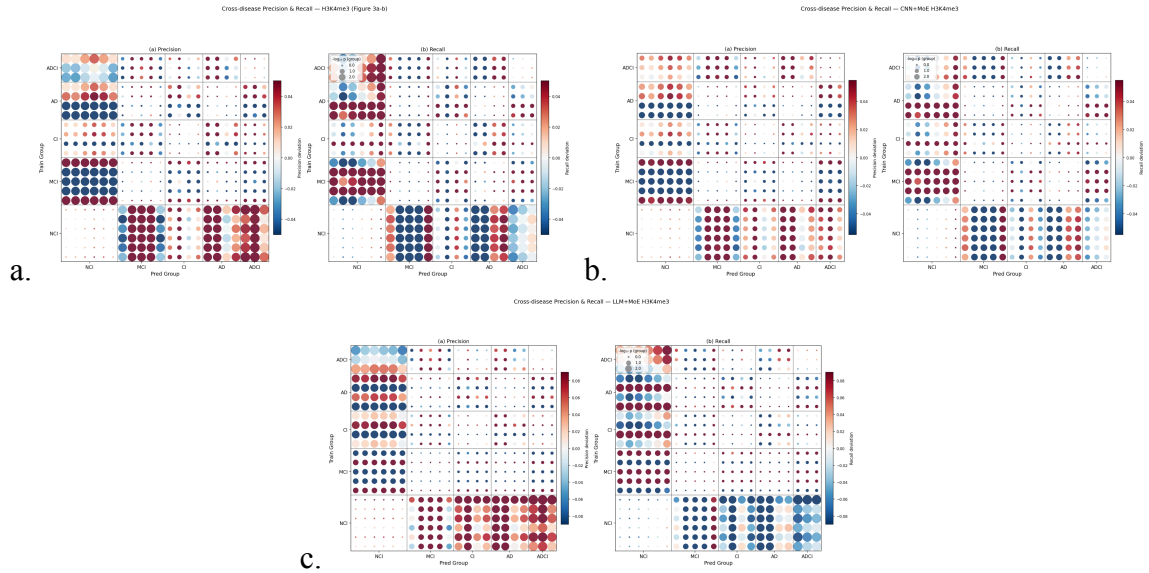
The performance gap between EpiModX and simpler baselines raises two questions: whether the gains are statistically meaningful, and whether the architectural complexity that produces them is warranted. To answer these, we evaluated two other model variants, a CNN+MoE reduced model and a cheaper CNN+BLSTM model, on the AD dataset across all three histone markers, allowing the contribution of each architecture component to be isolated.



AUROC of three reproduced models across 3 histone modifications in AD dataset

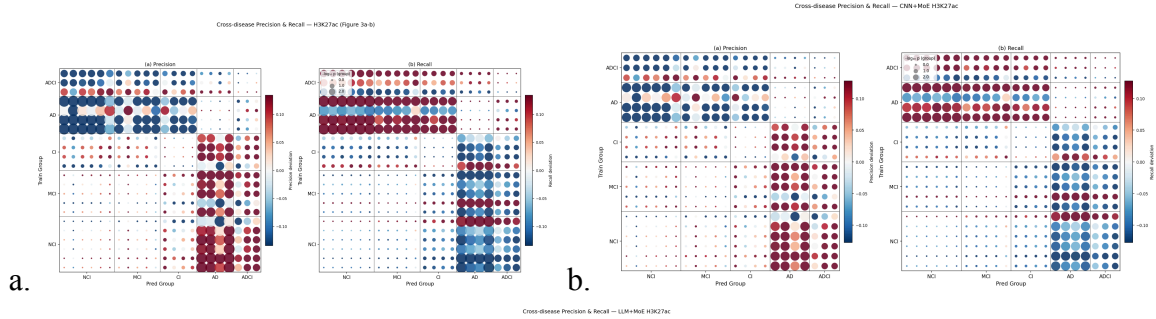
Across our reproduced models, AUROC values show minimal variation among three model architectures within each histone modifications, and the ranking also differs, suggesting that the performance gap between EpiModX and baseline models reported in the original benchmark may not be stable. Notably, LLM+MoE consistently achieves the lowest AUROC of the three architectures across all three histone modifications, contrary to expectations given its additional complexity. These results give rise to questions whether the architectural complexity of the full EpiModX pipeline translates to a meaningful and reliable performance advantage.

To further answer this question, we check the model prediction results and evaluate the model classification precision.



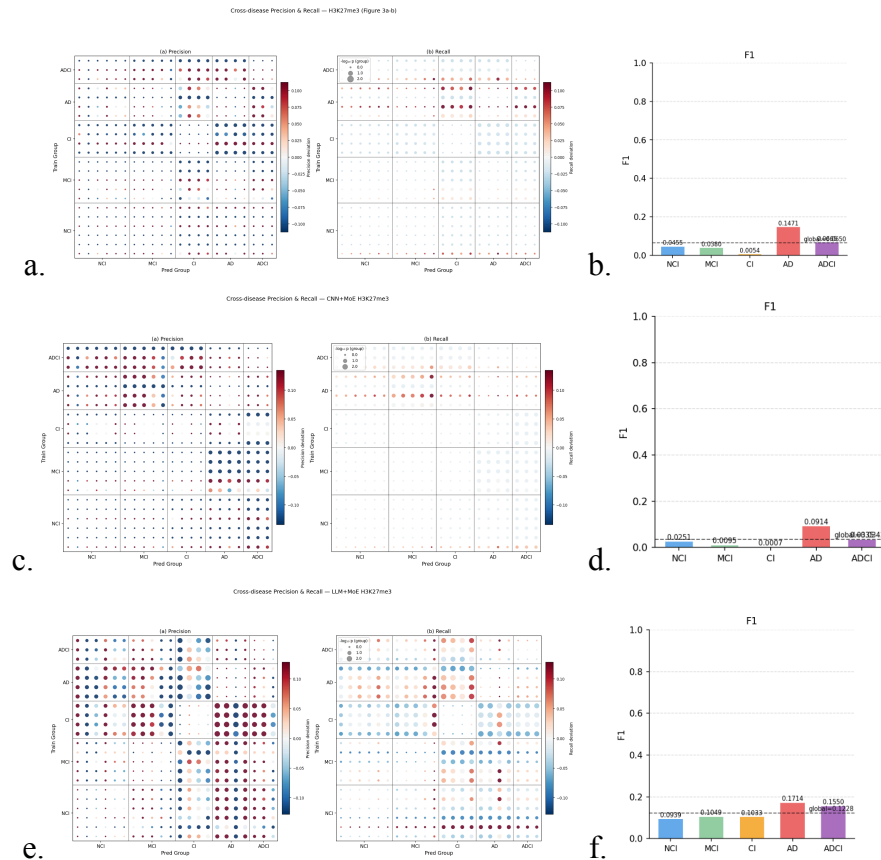
Precision and recall metrics of H3K4me3 across disease groups. a. CNN+BLSTM, b. CNN+MoE, c. LLM+MoE

For H3K4me3, all three model architectures produce consistent cross-patient transfer patterns. The NCI group shows the greatest precision asymmetry relative to MCI, CI, AD, and ADCI. Beyond the NCI separation, inter-group differences are small, and the overall similarity across the three models further supports that H3K4me3 prediction is robust to model architecture choice.



Precision and recall metrics of H3K27ac across disease groups. a.CNN+BLSTM, b.CNN+MoE, c.LLM+MoE

For H3K27ac, all three model architectures produce more consistent, and even identical cross-patient transfer patterns. The NCI, MCI groups show the precision asymmetry relative to CI, AD, and ADCI groups. Beyond this separation, inter-group differences seem neglectable.



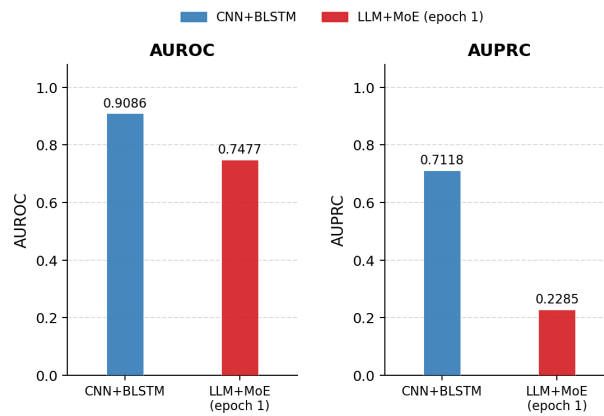
Precision and recall metrics of H3K27me3 across disease groups. a.CNN+BLSTM, c.CNN+MoE, e.LLM+MoE. Model F1 evaluation of b.CNN+BLSTM, d.CNN+MoE, f.LLM+MoE.

For the H3K27me3 training input being the largest matrix and the extra large number of training steps of H3K27me3, we only have fully trained CNN+BLSTM and LLM+MoE models and a 5-epoch test CNN+MoE model. The CNN+MoE shows worse model performance. It is reasoned in the original study that H3K27me3 shows consistency across all 5 groups. It is expected that all models show F1 collapse due to low-confident predictions (CNN+BLSTM: % predictions > 0.5: 0.7709485325393449 %, % predictions > 0.3: 16.183248907621515 %, % predictions > 0.1: 66.3940876222884 %, pred range: min=0.0001 max=0.7295 mean=0.1713; LLM+MoE: % predictions > 0.5: 1.73%, % predictions > 0.3: 7.46%, % predictions > 0.1: 50.75%, pred range: min=0.0015 max=0.8818 mean=0.1307). The inter-group patterns are similar between these three models except that AD, ACI groups are not as well-distinguished from NCI, MCI groups in CNN+BLSTM as in LLM+MoE. The marginal advantage suggests the LLM backbone extracts weak but meaningful signals that CNN misses, not the MoE structure.

Across all three histone modifications, the cross-patient precision pattern consistency reflects genuine biological signals captured rather than model-specific artifacts. The histone modifications differences among disease groups are learnable from sequence alone and are not much sensitive to architecture choice. H3K27me3 presents a harder prediction problem, where score compression and F1 collapse indicate that the models are under-confident. In this case, the complex LLM backbone has learned to be confident only on a small subset of obvious positive regions, which cannot be achieved using CNN.

3. AML Dataset Generalization:

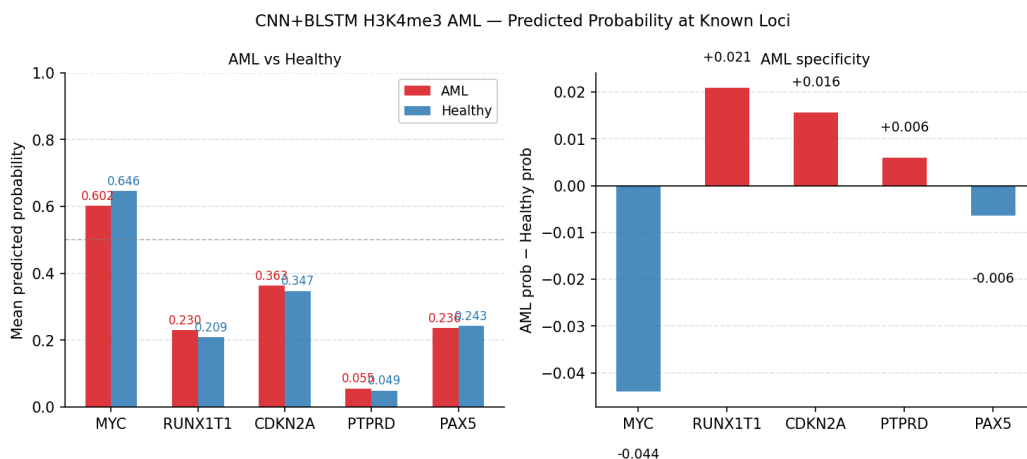
The AML dataset is formulated as a binary classification task, which is supposed to provide clearer class separability in histone modification patterns compared to the AD dataset, where disease progression is better characterized as continuous. We therefore use the AML task as a controlled setting to evaluate model generalization and to assess whether the added complexity of the model architecture yields worthy performance gains.



AUROC and AUPRC comparison between CNN+BLSTM and LLM+MoE_epoch1

For the time limitation, we are only able to generate a fully-trained CNN+BLSTM model and LLM-MoE model trained through one epoch. No useful conclusion can be drawn from this comparison.

Based on this, we want to further assess whether the CNN+BLSTM model captures biologically meaningful signals beyond aggregate metrics. We check the predicted probabilities at five genomic loci on the Chr8+Chr9 test chromosomes that have established links to H3K4me3 regulation and AML pathogenesis.



CNN+BLSTM Model Predicted Probability at Known Loci

At the MYC promoter, both AML (0.620) and healthy (0.646) groups receive high predictions with no AML-specific elevation. This finding is biologically correct because MYC carries constitutive H3K4me3 in all proliferating cells and its oncogenic contribution in AML arises through H3K27ac-dependent super-enhancer amplification rather than H3K4me3 gain.

At RUNX1, the t(8;21) translocation partner, the model correctly assigns higher probability to AML patients (0.230 vs 0.209), though the small margin reflects subtype heterogeneity across the 38 AML donors.

At CDKN2A, a modestly higher AML prediction (0.363 vs. 0.347) alongside a label rate of 0.40 reflects the biological complexity of this locus: while homozygous deletion silences it in ~15% of AML, the majority of patients retain an active, H3K4me3-marked allele. PTPRD shows uniformly low predictions in both groups, consistent with its AML relevance arising from copy number loss rather than epigenetic gain. PAX5, a B-cell lineage locus, shows no group difference as expected negative control result.

Overall, the directional agreement between model predictions and known AML biology at each locus suggests that the model has learned sequence features relevant to H3K4me3 deposition, even if the absolute difference remains modest given the heterogeneous multi-patient training signal.

Discussion

We were able to substantially reproduce the results of the original EpiModX study framework. Our results demonstrate that disease-specific histone modification patterns can be learned from sequence-based models, as our reproduced EpiModX pipeline uncovered prediction trends that broadly align with the original study. Despite these results, our ablation study suggests that the advantages of the full model architecture might have been overstated in the original benchmark. Across the AD dataset, we plotted AUROC differences between the full model, CNN+MoE, and CNN+BLSTM and in some cases the full model did not outperform the simpler alternatives. This result suggests that a large portion of the predictive signal may have already been captured by the simpler encoders.

It is also worth noting that the advantages of the DNA language model may strongly depend on the way the model is trained (e.g. number of epochs), the sequence length, and computational abilities. In our case, we reduced the input sequence length from 4096 to 2048 because of memory limitations, as well as increased batch size as needed to speed up training time.

The H3k27me3 histone was notably the most difficult prediction task, partly due to the scale of the data as well as its consistency among the five groups. This also potentially suggests that this particular histone may be more complex than sequence alone can tell us. For H3k27me3, the full model pipeline doesn't necessarily provide a dramatic performance gain on aggregate metrics, but it still outperforms other model architectures in capturing weaker disease-specific structure that the smaller models cannot discriminate. This could imply that the true benefit of LLM backbone may not lie in large performance improvements, but in its ability to handle more complex or region-specific epigenetic settings. In that regard, the computational cost of including a third model component may be difficult to justify for histones that already perform moderately with simpler methods, but may be more advantageous when the goal is to identify complex disease-specific structure that is proven difficult to detect.

Team Contributions

Alice:

- a. LLM + MoE data parallelization
- b. LLM + MoE training on AD datasets
- c. LLM + MoE generalization on AML dataset

Tiffany:

- d. LLM + MoE testing on AD dataset
- e. LLM + MoE testing on AML dataset
- f. Processing of outputted predictions + labels for generating metrics and plotting

Shunjia:

- g. Construction of AML dataset training input sequence
- h. CNN+BLSTM ablation study on AD dataset
- i. CNN+BLSTM generalization and Validation on AML dataset
- j. Figures generation

Max:

- k. CNN+MoE ablation study on AD dataset
- l. Collaborative slide/report preparation
- m. CNN+MoE on H3K4me3 for AML (currently still running on SCC... will update results accordingly)